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Mini Project Report

E-Ration Service Portal Analysis Using PowerBI

Bachelor of Technology (Computer Science (DS))

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E-RATION SERVICE PORTAL ANALYSIS USING POWER BI

1. Abstract

This project presents an analytical Power BI dashboard built on top of the ERation Service Portal a web-based government ration distribution system. The portal digitizes the Public Distribution System (PDS) by enabling citizens to check shop availability, stock levels, and queue count from home, while shopkeepers manage operations in real-time. The dataset covers 6 months (January to June 2026) across 10 ration shops and 100 registered citizens, containing 900 records spread across 7 tables: Admin, Users, Ration Shop, Ration Stock, Ration Distribution, Complaints, and Emergency Requests. Using Power BI, five interactive dashboards were developed to provide operational insights a master operations command center, stock monitoring, distribution analysis, complaints tracking, and emergency request management. These dashboards help government administrators and shop managers make data-driven decisions to improve ration distribution efficiency, reduce complaints, and ensure timely emergency response.

2. Objective

- To analyze ration distribution patterns across villages and months
- To monitor commodity-wise stock levels and identify shortage risks
- To track complaint resolution rates and identify problematic shops
- To assess emergency request approval trends and shop-wise demand
- To visualize citizen demographics including ration card type distribution
- To provide real-time queue and shop open/closed status insights
- To support government administrators with data-driven decision making

3. Dataset Description

The dataset is sourced from the E-Ration Service Portal database comprising 7 interconnected tables with a total of 900 records covering January to June 2026. Each table captures a distinct aspect of the ration distribution system.

Table 1: Database Schema Overview

Table Name	Records	Key Fields	Purpose
Admin	10	admin_id, username, password	Shopkeeper accounts
Users	100	user_id, ration_card_no, village, family_members, card_type	Citizen registrations
Ration_Shop	10	shop_id, shop_name, is_open, queue_count	Shop details & live status
Ration_Stock	60	shop_id, rice_qty, wheat_qty, sugar_qty, kerosene_qty, month	Monthly stock per shop
Ration_Distribution	600	user_id, shop_id, rice_given, wheat_given, village, month	Monthly distribution records
Complaints	80	complaint_id, shop_id, status, month	Citizen complaints

Emergency_Requests	50	request_id, user_id, shop_id, reason, status	Urgent supply requests

4. Data Preprocessing

- Imported all 7 Excel sheets into Power BI Desktop as separate tables
- Established relationships between tables using shop_id and user_id as keys
- Changed data types: month column set to Date, qty columns set to Whole Number, status fields set to Text
- Added a conditional column 'Shop Status' in Ration_Shop converting is_open (0/1) to 'Open'/'Closed' labels
- Extracted month name from distribution_date for time-series analysis
- Verified no null values or duplicates existed in primary key columns
- Created a unified date hierarchy for consistent month-wise filtering across all dashboards

5. Data Transformation & DAX Calculations

Several measures and calculated columns were created using DAX (Data Analysis Expressions) to derive meaningful KPIs and support dynamic dashboard visuals.

Calculated Columns Created

1. Shop Status

Converts the binary is_open field (0 or 1) into a readable 'Open' or 'Closed' label for display in visuals.

Formula:

Shop Status = IF(Ration_Shop[is_open] = 1, "Open", "Closed")

```
Shop Status = IF(Ration_Shop[is_open] = 1, "Open", "Closed")
```

2. Commodity Total (Stock)

Sums rice_qty, wheat_qty, sugar_qty, and kerosene_qty to calculate total stock available at each shop per month

```
Commodity Total =
SUM(Ration_Stock[kerosene_qty]) +
SUM(Ration_Stock[sugar_qty]) +
SUM(Ration_Stock[wheat_qty]) +
SUM(Ration_Stock[rice_qty])
```

Measures Created

1.Total Users - Count of distinct users registered in the system.

```
Total Users = COUNTROWS(Users)
```

2. Total Distributions - Count of all distribution records across all months.

```
Total Distributions = COUNTROWS(Ration_Distribution)
```

3. Total Shops - Count of distinct shops in the Ration_Shop table**.**

```
Total Shops = COUNTROWS(Ration_Shop)
```

4. Pending Complaints - Count of complaints where status equals 'Pending'.

```
Pending Complaints = CALCULATE(COUNTROWS(Complaints), Complaints[status] = "Pending")
```

5. Resolved Complaints - Count of complaints where status equals 'Resolved'.

```
Resolved Complaints = CALCULATE(COUNTROWS(Complaints), Complaints[status] = "Resolved")
```

6. Resolution Rate % - Percentage of complaints that have been resolved out of total complaints.

```
Resolution Rate % = DIVIDE([Resolved Complaints], COUNTROWS(Complaints), 0) * 100
```

7. Total Different Types of Card Users- Count of different type of card user

```
AAY Users = CALCULATE(COUNTROWS(Users), Users[card_type] = "AAY")
```

```
APL Users = CALCULATE(COUNTROWS(Users), Users[card_type] = "APL")
```

```
BPL Users = CALCULATE(COUNTROWS(Users), Users[card_type] = "BPL")
```

8. Total Rice/Wheat/Sugar/Kerosene Distributed - Sum of respective given columns from Ration_Distribution table.

```
Total Rice Distributed = SUM(Ration_Distribution[rice_given])
```

```
Total Sugar Distributed = SUM(Ration_Distribution[sugar_given])
```

```
Total Wheat Distributed = SUM(Ration_Distribution[wheat_given])
```

```
Total Kerosene Distributed = SUM(Ration_Distribution[kerosene_given])
```

9. Approved Requests - Count of emergency requests where status equals 'Approved'.

```
Approved Requests = CALCULATE(COUNTROWS(Emergency_Requests), Emergency_Requests[status] = "Approved")
```

10. Avg Queue Count – Average of queue_count in overall shops.

```
Avg Queue Count = AVERAGE(Ration_Shop[queue_count])
```

Purpose of DAX Calculations

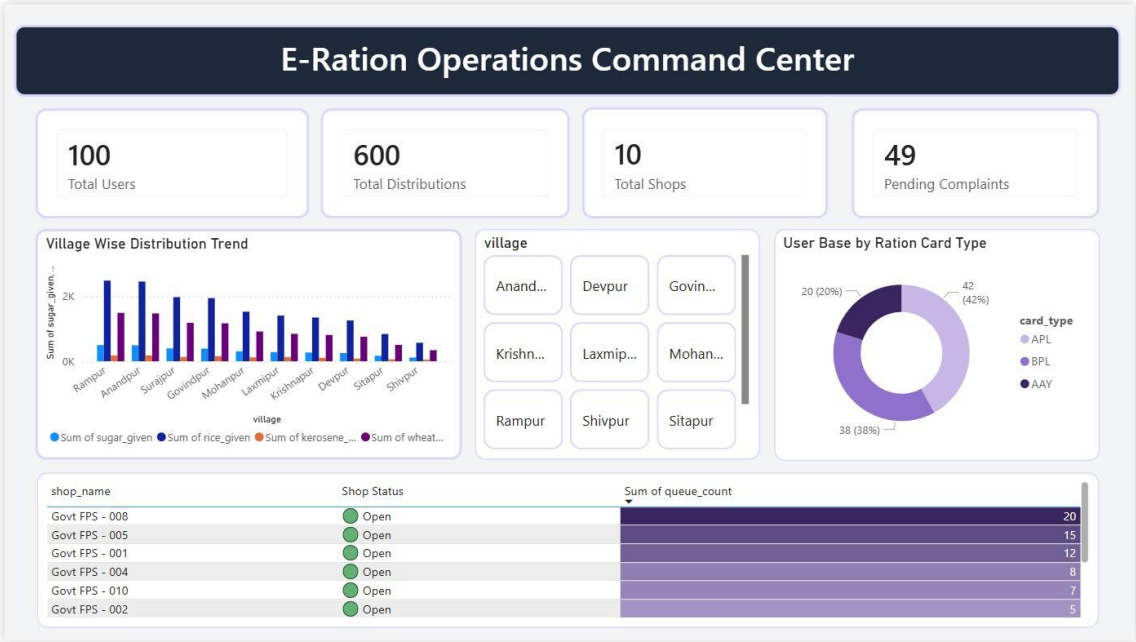
- To convert raw database fields into actionable KPI metrics
- To enable dynamic filtering across all 5 dashboards using slicers
- To compute aggregated totals for stock, distribution, and complaint tracking
- To support visual formatting with conditional colors based on status values

6. Dashboard Explanation

Dashboard 1: E-Ration Operations Command Center (Master Dashboard)

This is the master overview dashboard providing a high-level summary of the entire E-Ration portal operations. It serves as the primary landing page for administrators to get a complete picture at a glance.

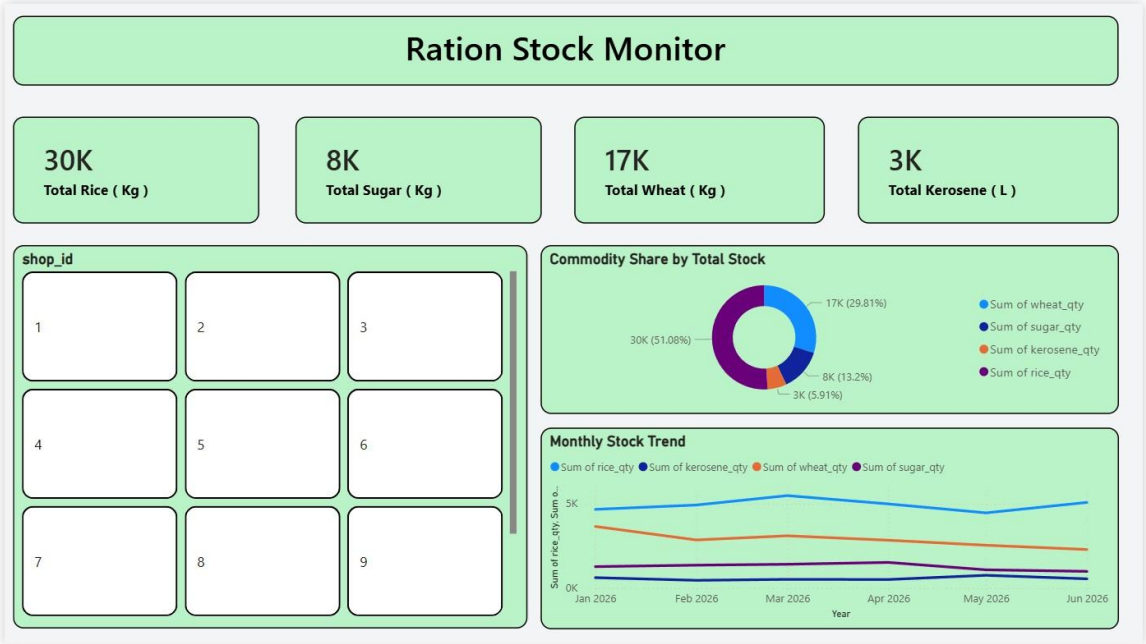
- KPI Cards display Total Users (100), Total Distributions (600), Total Shops (10), and Pending Complaints (49)
- Village Wise Distribution Trend - Clustered bar chart showing rice, wheat, sugar, and kerosene distributed per village, enabling comparison across all 10 villages
- Village Slicer - Tile-style filter allowing selection of one or more villages to drill down into village-specific data across all visuals
- User Base by Ration Card Type - Donut chart showing the split between APL (42%), BPL (38%), and AAY (20%) card holders
- Shop Queue & Status Table - A data bar table listing open shops ranked by queue count, helping administrators identify shops with high footfall requiring management attention



Dashboard 2: Ration Stock Monitor

This dashboard focuses exclusively on commodity stock levels across all shops and months. It enables supply chain monitoring and identification of stock shortages before they become critical.

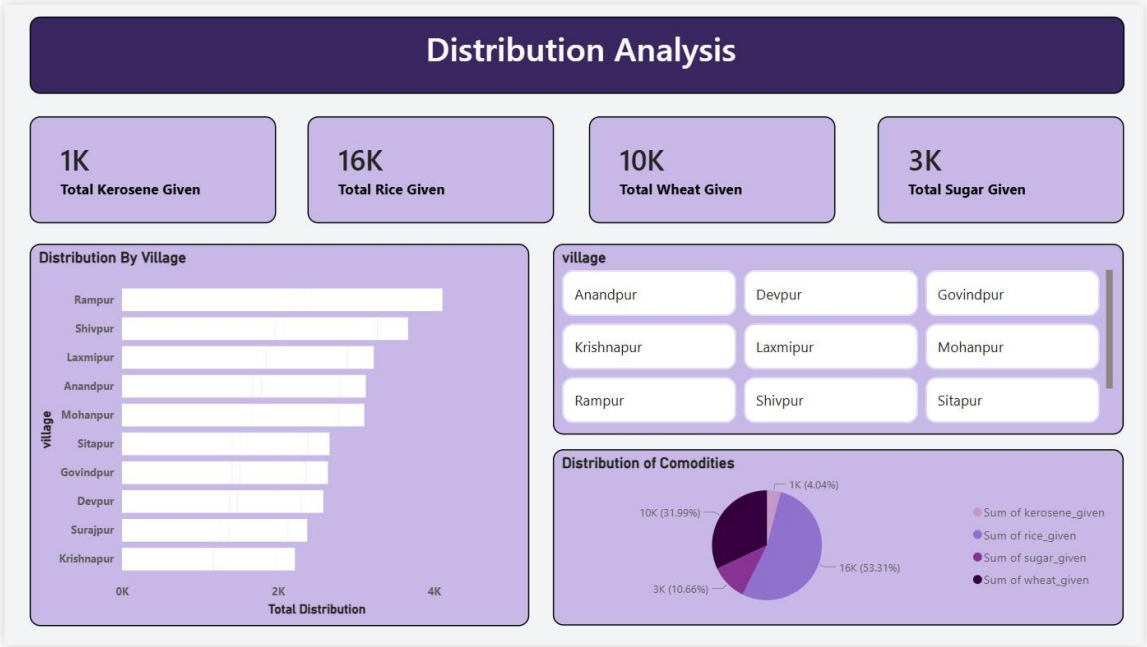
- KPI Cards show Total Rice (30K kg), Total Sugar (8K kg), Total Wheat (17K kg), and Total Kerosene (3K litres) across all shops
- Shop ID Slicer - Tile-based slicer allowing selection of individual shops to view their specific stock data
- Commodity Share by Total Stock - Donut chart showing the proportion of each commodity in total stock: Rice (51%), Wheat (30%), Sugar (13%), Kerosene (6%)
- Monthly Stock Trend - Line chart tracking stock levels of all four commodities across January to June 2026, showing how inventory changes over time and identifying months with notable dips or surges



Dashboard 3: Distribution Analysis

This dashboard provides a detailed breakdown of ration distribution activities, helping identify which villages and commodities have the highest demand and ensuring equitable distribution across the region.

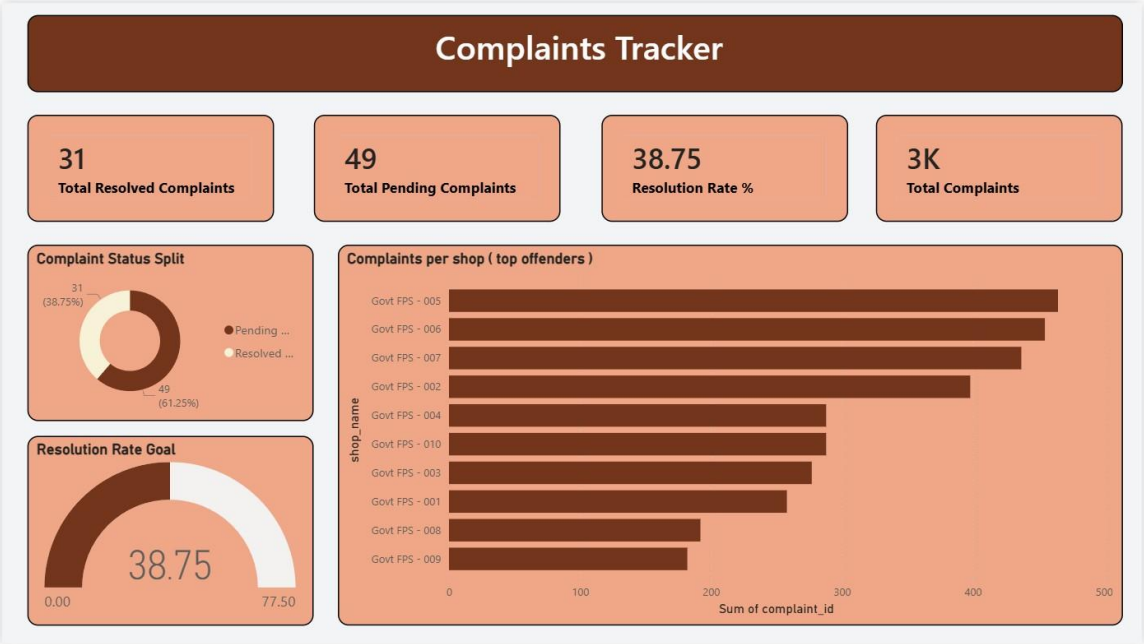
- KPI Cards show Total Kerosene Given (1K L), Total Rice Given (16K kg), Total Wheat Given (10K kg), and Total Sugar Given (3K kg)
- Distribution By Village - Horizontal bar chart ranking all 10 villages by total ration distributed, with Rampur and Shivpur leading, enabling identification of high-demand zones
- Village Slicer - Tile filter to drill into a specific village's distribution data
- Distribution of Commodities - Pie chart showing the share of each commodity in total distribution: Rice (53%), Wheat (32%), Sugar (11%), Kerosene (4%)



Dashboard 4: Complaints Tracker

This dashboard monitors citizen grievances and complaint resolution efficiency. It helps government administrators identify shops generating the most complaints and evaluate shopkeeper performance.

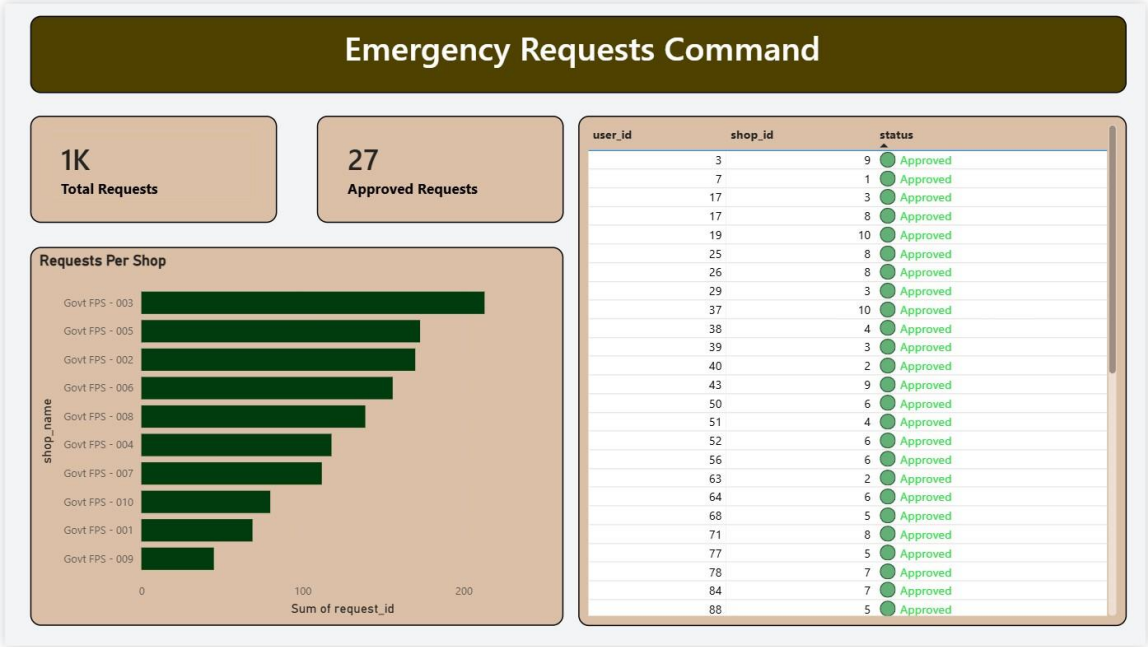
- KPI Cards show Total Resolved Complaints (31), Total Pending Complaints (49), Resolution Rate % (38.75%), and Total Complaints (80)
- Complaint Status Split - Donut chart showing Pending (61.25%) vs Resolved (38.75%) complaints, indicating that over half of complaints remain unresolved
- Resolution Rate Goal - Gauge visual displaying current resolution rate of 38.75 against a target of 77.50, highlighting the need for improved complaint handling
- Complaints Per Shop (Top Offenders) - Horizontal bar chart ranking shops by complaint volume, identifying Govt FPS-005, FPS-006, and FPS-007 as the most complained-about shops requiring immediate attention



Dashboard 5: Emergency Requests Command

This dashboard tracks urgent supply requests submitted by citizens facing exceptional circumstances such as natural disasters, medical emergencies, or loss of food stock.

- KPI Cards display Total Requests (1K) and Approved Requests (27)
- Requests Per Shop - Horizontal bar chart showing which shops received the most emergency requests, with Govt FPS-003 and FPS-005 receiving the highest volume
- Approved Requests Table - Detailed table showing user_id, shop_id, and status for all approved emergency requests, enabling administrators to track who has received emergency ration support



7. Insights

- APL cardholders form the largest segment (42%) followed by BPL (38%) and AAY (20%), indicating a predominantly above-povertyline registered user base
- Rice accounts for over 50% of total stock and 53% of total distribution, making it the most critical commodity to monitor for shortages
- Rampur and Shivpur villages consistently show the highest ration distribution volumes, suggesting higher population density or family sizes in these areas
- The complaint resolution rate stands at only 38.75%, with 49 out of 80 complaints still pending, indicating a significant backlog in grievance redressal
- Govt FPS-005, FPS-006, and FPS-007 are the top complaintgenerating shops, suggesting operational issues or service quality problems at these locations
- Shop queue counts range from 3 to 20 at open shops, with Govt FPS008 having the longest queue, indicating uneven footfall distribution

- Monthly stock trends remain relatively stable across January to June 2026 with slight dips in April and May, suggesting seasonal demand fluctuations
- Emergency requests show higher concentration at FPS-003 and FPS005, which may correlate with higher poverty levels or disaster-prone areas in those villages
- 3 shops (FPS-003, FPS-006, FPS-009) were found closed during data collection, affecting citizen access and potentially contributing to emergency requests

8. Conclusion

This project demonstrates how Power BI can be effectively applied to a realworld government service system to extract actionable insights from operational data. The five dashboards built on the E-Ration Service Portal dataset provide a comprehensive view of ration distribution operations from high-level KPIs to granular shop-level performance. The analysis reveals key gaps in complaint resolution, stock monitoring needs, and uneven distribution patterns across villages. By leveraging DAX measures, interactive slicers, and well-designed visualizations, this Power BI solution enables government administrators to monitor the public distribution system efficiently, identify problem areas proactively, and take corrective action to ensure fair and timely ration delivery to all registered citizens.

9. Future Scope

- Integration of real-time data from the live portal database using Power BI's DirectQuery or scheduled refresh
- Predictive analytics using machine learning to forecast stock requirements and prevent shortages
- Automated alert system in Power BI to notify administrators when stock falls below threshold or complaints exceed target
- Geospatial mapping using Power BI's map visuals to show villagewise distribution on an actual map

- Addition of citizen satisfaction scores and feedback ratings as new data dimensions

10.References :

Rishabsarda/erationservice_portal: A portal for citizen to check the stock availability and crowd at ration shop so that they don't need to wait in line for ration and no time is wasted

- Mobile-optimized Power BI report layout for field-level shopkeeper monitoring

- Integration with Aadhaar-based fingerprint verification data to track authentic collection rates